

Problem and research questions

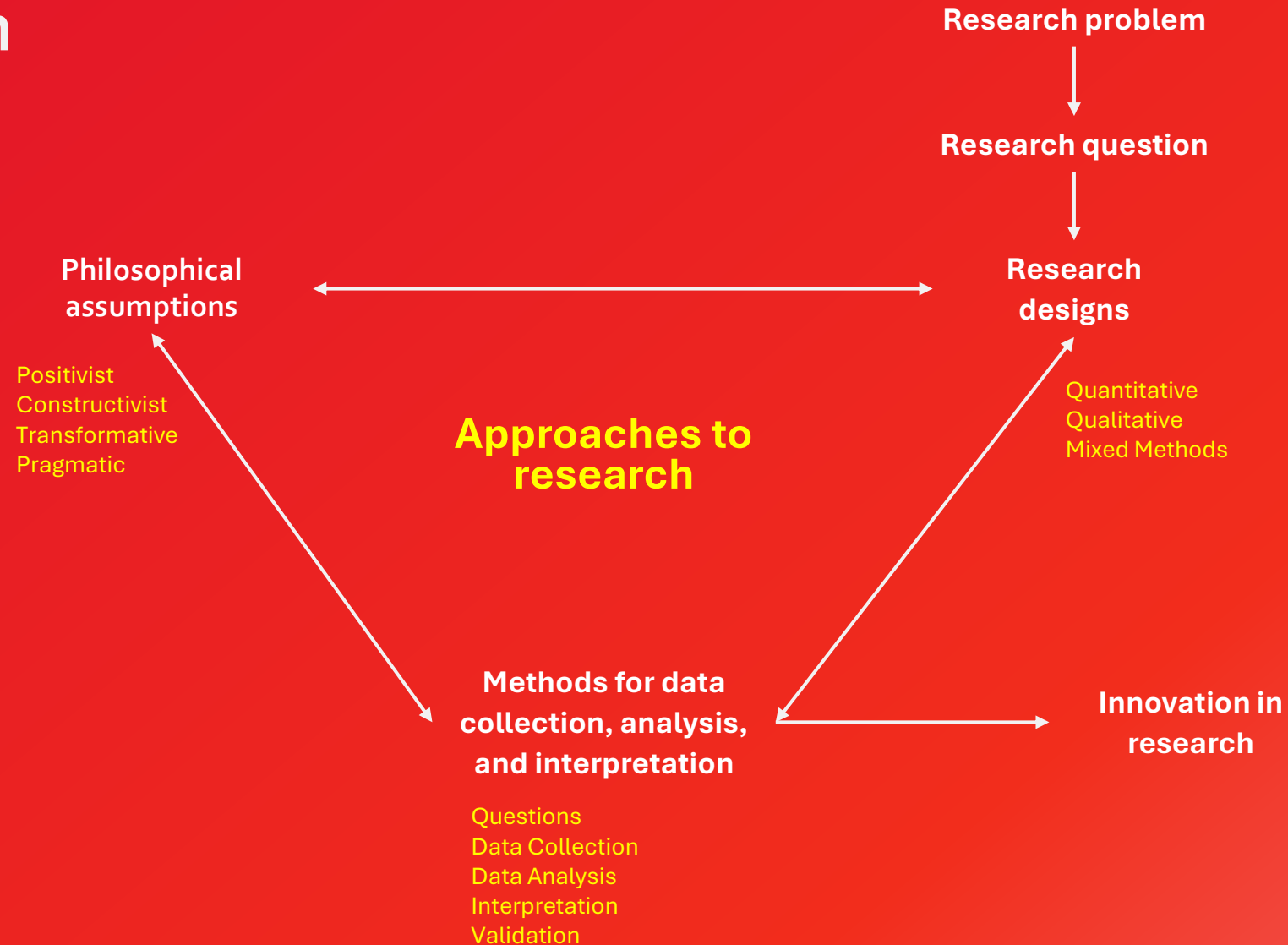


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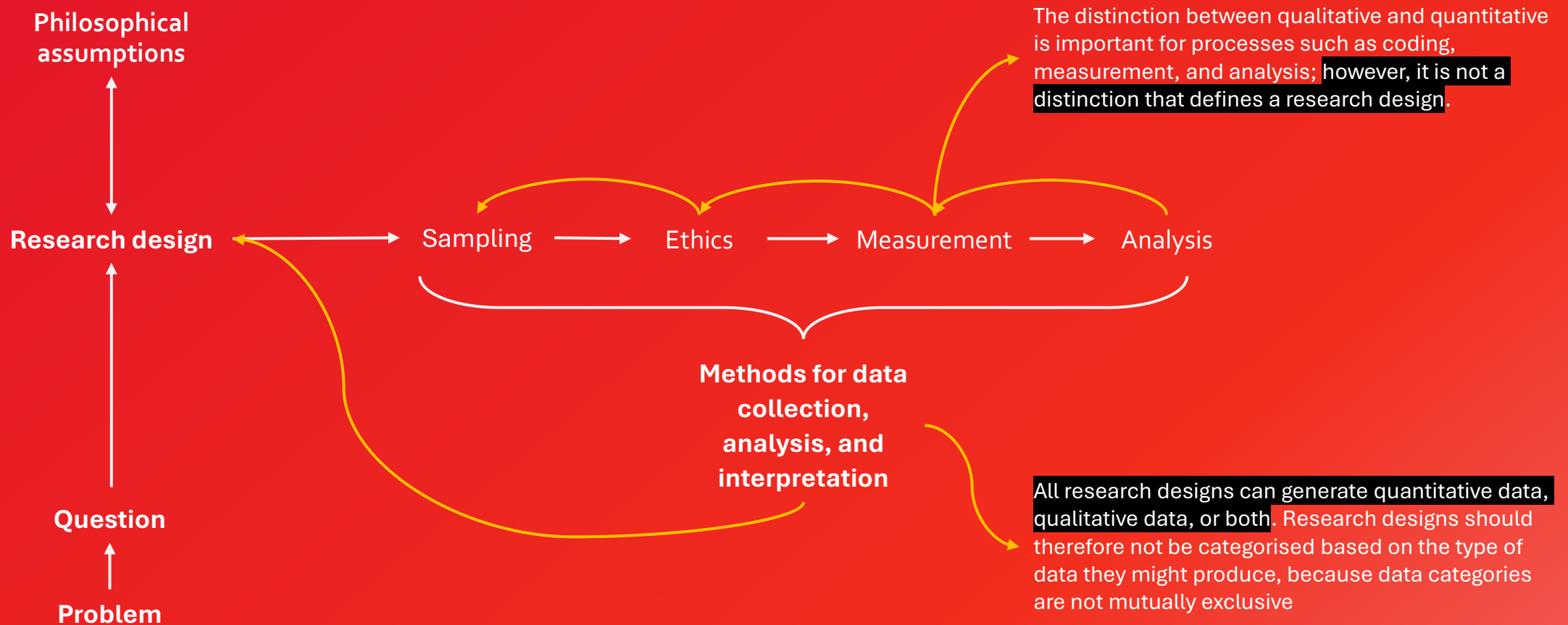
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A Framework for Research



A Framework for Research, from the problem



How to select the right method?

The question “Which method?” is particularly important when selecting research designs, because **the choice of design significantly influences most of the subsequent decisions a researcher must make.**

✓ Focus on the problem

Research designs are most effective, and have the greatest potential for innovation, when they are determined by the nature of the research problem.

- This choice is central to innovation because, if researchers do not approach it consciously, **they may simply maintain the status quo of an existing paradigm**, following established traditions without making deliberate methodological decisions.
- Therefore, you should **actively seek the most effective methods**, rather than simply following conventional or pre-existing methodological paths.

How to select the right method?

There's non-exclusivity of Methods within Research Designs

Surveys



Quant

Survey design requires multiple iterations of question phrasing and structural refinement, particularly about the audience's comprehension of the content and purpose of each question.



This is true even for items whose responses are ultimately numerically coded.

Although the analysis of survey data is frequently quantitative, it is not exclusively so. **Surveys may be used to collect qualitative, quantitative, or mixed forms of evidence**

Design BEFORE methods

1. The Design Shapes the Method

Research design informs the development of methodological procedures, including:

- Sample selection
- Ethical considerations
- The construction of instruments or strategies for data collection.

2. Changing the design to accommodate the methods is costly

Changing the design is costly. **Changing the design requires starting again from the beginning**, as it determines the entire structure of the research process.

How to select the right method?

Question

Which are the main factors driving the dropout rates in the university?

- Which type of knowledge is needed to address this question?

Problem

Students are leaving university early, but why they leave is not clearly understood

- What is the essence of the problem?



Design

Mixed methods, sequential explanatory design to:

- Understanding students lived experiences
- Determining which factors statistically predict dropout



Sample

- Who, where, when?

Ethics

Measurement

- How?

Analysis



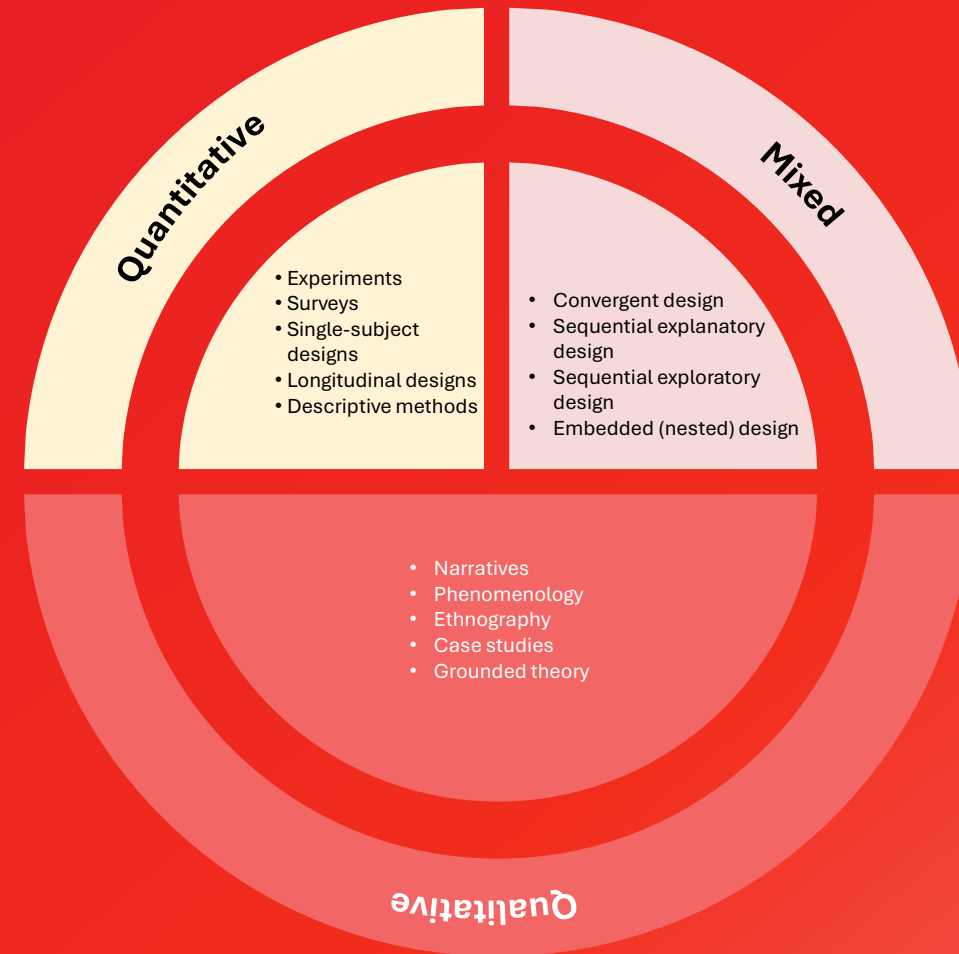
Methods

**So, lets move from a problem,
to a question to a design**

Research designs

Within the available methods (quantitative, qualitative, and mixed methods), **YOU, as a researcher, must select the specific designs used to develop the project.**

Research designs are types of inquiry within methodological approaches that provide a specific direction for the procedures to be carried out in a study.



Research designs

- Research designs require the researcher to **choose, from among available quantitative, qualitative, and mixed-methods approaches, the specific design that will be used to develop the project.**
- Research designs are types of inquiry within methodological approaches that provide a specific direction for the procedures to be carried out in a study.

“A research design is a plan for collecting evidence that can be used to answer a research question.”

- **A sound research design must be justified in relation to the research question.**
- At a minimum, the researcher should provide a clear rationale demonstrating that the chosen design is effective for addressing the task posed by the question.

Problem, question and design

How do we move from a research question and problem to a research design?

To address a research question, YOU must determine whether the study requires:	Yes	No	
Causal inference			} Nature of the answers needed to address the research problem.
Generalization from the cases studied (sample) to a wider population			
Analysis of change over time			
Interaction with participants			} Nature of the data or sources of evidence required to address the problem.
Generation of primary data			
The use of more than one design			

Decisions about how to code and analyse the evidence, that is, decisions about methods, must not determine the answers to these questions.

If this happens, **the researcher risks confusing the aim of answering the research question with the nature of the evidence used to answer it.**

Problem, question and design

How do we move from a research question and problem to a research design?

1	Ask yourself:		Yes	No
	To answer the research question, is there need to make Causal inferences ?			
✓	Because justified causal inferences arise more from the design than from measurement or analysis, it is crucial to incorporate causal considerations at the early stages of planning a research project.			

Any effort to relate one thing to another (one variable to another, or one phenomenon to another) is ultimately causal.

- There are multiple methods for making causal inferences (e.g., experiments and quasi-experiments).
- A causal approach does not exclude non-causal descriptive elements within a piece of research.

Problem, question and design

How do we move from a research question and problem to a research design?

2

Ask yourself:

To answer the research question, is there need **to generalise from the cases studied (the sample) to a wider group (the population)**?

Yes

No

It is essential to determine whether the primary interest lies in the specific cases within the sample, or whether the intention is to generalise.

- Although generalisation is not a purely dichotomous decision (since any processing of results entails some degree of generalization) clarity on this point is necessary.



When the principal aim of the research is to generalise, particular attention must be paid to selecting a sample from which legitimate generalisation is possible (e.g., random, stratified, or probability-based sampling)



When the principal aim is not to generalise, as in case studies, generalisation may instead be employed as a tool to help describe and explain the specific case.

Problem, question and design

How do we move from a research question and problem to a research design?

3	Ask yourself:	Yes	No
	To answer the research question, is there need to examine change over time ?		

It is necessary to determine whether the interest lies in studying:

- Individuals, cases, or variables over time
- Or whether a cross-sectional approach is sufficient.

- ✓ **In experimental settings**, the time that elapses between the treatment and the measurement becomes a confounding variable, producing “history effects”.
- ✓ **In prospective studies** or retrospective historical studies, time is the central focus of the research.

Problem, question and design

How do we move from a research question and problem to a research design?

4	Ask yourself:	Yes	No
	To answer the research question, is there need to interact with the subjects/participants?		

It is necessary to determine whether direct interaction with the objects of study will provide information that could not be obtained otherwise.

- Interaction may also bias the results and the data-generation process.
- ✓ **In both medical and social research, a pluralistic approach is often the most effective;** combining perspectives from different data sources can be an important means of generating new insights and methodological innovations.

Problem, question and design

How do we move from a research question and problem to a research design?

5	Ask yourself:	Yes	No
	To answer the research question, is there need to generate your own sources and data?		

- ✓ **Data sources:** The question concerns whether it is necessary to produce original data or whether publicly available records or archival datasets can be used
- ✓ **Types of archival research:** Examples include historical research and meta-analyses that rely on secondary sources.

Determining the sources of data is crucial, as **in some cases no alternative exists to answer the research question other than generating one's own evidence.**

Problem, question and design

How do we move from a research question and problem to a research design?

6	Ask yourself:	Yes	No
	To answer the research question, is there need use more than one design?		

- ✓ **It is necessary to determine whether the costs of employing more than one design are outweighed by the benefits of doing so:** particularly when the use of multiple designs may combine their strengths and mitigate the weaknesses of single-design approaches.

Problem, question and design

“Evaluating the six questions (proposed by Vogt) helps to clarify the available design options.

Designing a research project involves a long journey across somewhat unfamiliar terrain and requires many decisions. These decisions must be made at an early stage”

Problem, question and design

1. Identify three topics
2. Develop the identification-of-design matrix: https://uexternadoedu-my.sharepoint.com/:w:/g/personal/luis_guevara_uexternado_edu_co/IQBNpEtpRTa9T79RVke-3aOBACq_WbM3ciuQ-ungugUZkjA?e=wkO88d
3. **Select one main topic**, based on the feasibility of implementing the design and its capacity to answer the research question.